# Akshay Patil

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## EDUCATION

**Georgia Institute of Technology** – Atlanta, GA **August 2026 – May 2029**
Bachelor of Science in Aerospace Engineering | Sophomore | GPA: 4.00
*Current coursework:* CS 1371 Computing, MATH 2551 Multivariable Calculus, COE 2001 Statics, ME 1670 Graphics, AE 1601 Aerospace Engineering, MSE 2001 Materials

**School of Science and Engineering at Townview** – Dallas, TX **August 2022 – May 2026**
3.97 Unweighted GPA | Rank 2 | 1600 SAT | National Merit Scholar | NASA High School Aerospace Scholar

## EXPERIENCE

**Hardpoint Robotics** | Co-Founder **June 2026 – Present**
- Hardpoint builds NDAA-compliant development UAVs, aircraft for customers who are not allowed to fly foreign-sourced hardware
- Designing airframes in CAD and 3D printing CF-Nylon parts for them
- Working with various North American vendors to manufacture the carbon fiber structures and source compliant components, where lead times shape the design as much as the loads do

**Salvo Dynamics** | Design Engineer **August 2026 – Present**
- Salvo builds small interceptor aircraft meant to fly out and bring down hostile drones, and is separately building an RC jet to break the speed of sound
- Turning optimized aerobody models into load-bearing structures in CAD, and laying out the avionics that go inside them
- Designing 3D printed airframes for testing, plus the assembly jigs the production aircraft are built on

**SWYFT Robotics** | Engineering Design Intern **May 2025 – May 2026**
- SWYFT sells competition drivetrain and structural parts to FIRST Tech Challenge and FIRST Robotics Competition teams
- Designed sprockets, gears, and pulleys that hold up under competition loads at a price teams can afford

## PROJECTS

**Autonomous House-Mapping Robot** | Personal Project **2025 – Present**
- Jetson Orin Nano robot that explores a house with no operator and counts windows, using an MPPI controller, RTAB-Map visual SLAM, and frontier exploration I wrote
- No lidar: obstacles come from monocular depth classified against a fitted ground plane, windows from three open-vocabulary detectors filtered by a vision-language model

**Pickleball Ball Boy UAV** | Personal Project **May 2025 – Present**
- Benchmarked five ball detection and four localization methods; AprilTag localization was the most accurate by a wide margin
- Split the drag model into horizontal and vertical coefficients, bringing landing prediction inside a 20 cm zone

## RESEARCH AND CLUBS

**Hypersonics VIP, Georgia Tech** | Data and Communications Team **August 2026 – Present**
- Building a sonic fatigue test facility, where acoustic loading is used to find how structures fail under sustained high-intensity sound
- Designing the sensor package that measures pressure, temperature, and airflow inside the facility

**FIRST Tech Challenge Team 7172** | Captain and Lead Designer **September 2021 – May 2026**
- Lead designer on four robots; 2026 World Championship Finalist Alliance, 19 event wins and 63 awards over five seasons
- Designed a shooter that sorts and fires three balls in 0.2 to 0.5 s using a wedge spindexer and a flywheel on a 200 Hz servo turret

## SKILLS

**Robotics:** ROS 2 (Nav2), RTAB-Map visual SLAM, MPPI control, AprilTag localization
**Programming:** Python (NumPy, Pandas, TensorFlow), MATLAB, JavaScript, SQL, Git
**Hardware:** NVIDIA Jetson, Raspberry Pi, Arduino, REV Control Hub
**CAD:** SolidWorks, Fusion 360, Onshape, Autodesk Inventor, 3D printing
**Certifications:** FAA Part 107 Remote Pilot, Autodesk Certified User (Inventor)